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DETACHABLE FIBER ARRAY UNIT FITTING WITH GUIDE PINS

Abstract

A structure comprises a ferrule receptacle, a device die extending into a cavity of the ferrule receptacle from a first edge of the ferrule receptacle, a guide pin at least partially in the cavity, and a detachable fiber array unit extending into the cavity from a second edge of the ferrule receptacle. The first edge and the second edge are opposite edges of the ferrule receptacle. The guide pin has a portion in a pin hole of the detachable fiber array unit.

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Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE [0001] This application claims the benefit of the following provisionally filed U.S. Patent application: Application No. 63/558,970, filed on Feb. 28, 2024, and entitled “DETACHABLE IFAU WITH GUIDE PIN,” which application is hereby incorporated herein by reference.

BACKGROUND

[0002] As the bandwidth requirement grows rapidly for high-performance computing systems, high-speed optical Input/Output (I/O) modules have been used increasingly. The optical I/O modules are often connected to light sources (laser) as the circuit driving sources.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1A, 1B, 1C, 1D, 1E, and 1F illustrate the views of a detachable fiber array unit and a photonic receiving unit including guide pins in accordance with some embodiments.

[0005] FIGS. 2A, 2B, 2C, and 2D illustrate the views of a detachable fiber array unit and a photonic receiving unit including guide pins in accordance with some embodiments.

[0006] FIGS. 3A, 3B, 3C, and 3D illustrate the views of a detachable fiber array unit and a photonic receiving unit including guide pins and alignment features in accordance with some embodiments.

[0007] FIGS. 4A, 4B, 5A, 5B, 6A, and 6B illustrate the views of detachable fiber array units and photonic receiving units in accordance with alternative embodiments.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “underlying,” “below,” “lower,” “overlying,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] A package including a detachable fiber array unit, a photonic receiving unit, and alignment features are provided. In accordance with some embodiments of the present disclosure, the package includes a detachable fiber array unit including pin holes, and a photonic receiving unit that further includes a ferrule receptacle including guide pins. The guide pins and pin holes are used as the

alignment features. The detachable fiber array unit is inserted into the ferrule receptacle, and the guide pins and the pin holes are used to align the detachable fiber array unit to the ferrule receptacle. Other alignment features including spring pushes may be used for further limiting and aligning the detachable fiber array unit to the ferrule receptacle. Additional alignment features having tilt surfaces are also used to limit and align the detachable fiber array unit to the ferrule receptacle.

[0011] Embodiments discussed herein are to provide examples to enable making or using the subject matter of this disclosure, and a person having ordinary skill in the art will readily understand modifications that can be made while remaining within contemplated scopes of different embodiments. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements. Although method embodiments may be discussed as being performed in a particular order, other method embodiments may be performed in any logical order.

[0012] Referring to FIGS. **1A**, **1B**, **1C**, **1D**, **1E**, and **1F**, a part of a photonic receiving unit **10** is illustrated, which includes an optical engine therein. In accordance with some embodiments of the present disclosure, package component **20** is provided for attaching other components thereon. Package component **20** may include a package substrate, a printed circuit board, a package including other package components such as device dies, or the like. Electrical conductive features (not shown) may be formed on the opposite sides of package component **20**, and are interconnected through the conductive paths (such as metal lines, vias, or the like, not shown) inside package component **20**.

[0013] In accordance with some embodiments of the present disclosure, package component **24** is placed over and bonding to package component **20**. The bonding may be performed through solder bonding, metal-to-metal direct bonding, dielectric-to-dielectric bonding, and/or the like. For example, solder bonding may be used for the bonding and electrically connecting package component **24** to package component **20**.

[0014] In accordance with some embodiments of the present disclosure, package component **24** comprises an optical engine, and thus may alternatively be referred to as optical engine **24**. In accordance with some embodiments of the present disclosure, package component **24** comprises a photonic Integrated circuit (PIC) die and an Electronic Integrated circuit (EIC) die bonding to the PIC die, which are not illustrated separately. The bonding between the EIC die and the PIC die may also include metal-to-metal direct bonding, solder bonding, or hybrid bonding.

[0015] In accordance with some embodiments of the present disclosure, the PIC die may include optical components including, and not limited to, silicon waveguides, (silicon) nitride waveguides, grating couplers, photodetectors, modulators, edge couplers, and/or the like. The PIC die may include a silicon substrate, on which the optical components are formed.

[0016] In accordance with some embodiments of the present disclosure, the EIC die may include integrated circuits for interfacing with the PIC, such as the circuits for controlling the operation of the PIC. For example, the EIC may include controllers, drivers, amplifiers, the like, or combinations thereof. the EIC may also include a CPU. In accordance with some embodiments of the present disclosure, the EIC includes the circuits for processing electrical signals received from the PIC. The EIC may also control high-frequency signaling of the PIC according to electrical signals (digital or analog) received from another device or die, in accordance with some embodiments. In accordance with some embodiments of the present disclosure, the EIC may include a circuit that provides Serializer/Deserializer (SerDes) functionality. In this manner, the EIC may act as a part of an I/O interface between optical signals and electrical signals.

[0017] Package component **26** is over and bonded to package component **24**. In accordance with some embodiments of the present disclosure, device die **26** is a die including a wave guide to optically couple to a Fiber Array Unit (FAU). In accordance with some embodiments of the present disclosure, package component **26** is a device die, and may be a PIC die, and may be a part of the

optical engine. Package component **26** may have the function of receiving optical signals or transmitting optical signals from/to a detachable fiber array unit, and receiving/transmitting the optical signals to package component **24**. In accordance with alternative embodiments, package component **26**, in addition to receiving or transmitting optical signal, may convert the optical signals to electrical signals or electrical signals to optical signal.

[0018] Ferrule receptacle **28** is attached to package components **26** and **24**. Ferrule receptacle **28** has a cavity **30** therein, so that ferrule **42** and ferrule housing **44** may be inserted into the cavity **30** of ferrule receptacle **28**.

[0019] In accordance with some embodiments of the present disclosure, grooves **32** are formed in package component **26**. Referring to FIG. **1E**, when package component **26** is a device die, the top surface portion of the package component **26** may be semiconductor substrate **27**. The semiconductor substrate **27** may further be a silicon substrate in accordance with some embodiments. Grooves **32** may extend from the top surface of the semiconductor substrate **27** into the semiconductor substrate **27**. Grooves **32** may be formed by etching, milling, sawing, or the like. The lengthwise direction of grooves **32** is parallel to the insertion (and pulling out) direction of detachable fiber array unit **40**, as will be discussed in detail in subsequent paragraphs.

[0020] As shown in FIGS. **1A** and **1F**, guide pins **36** are placed in, and thus are fixed by, grooves **32**. In accordance with some embodiments, the lower parts of guide pins **36** are in grooves **32**, while the upper parts of guide pins **36** are out of grooves **32**. Top lid **34** is placed on and pressed on guide pins **36**, so that guide pins **36** are secured by top lid **34**. In accordance with some embodiments as shown in FIG. **1E**, adhesive **33** may be dispensed in grooves **32** to further secure guide pins **36** in positions.

[0021] In accordance with some embodiments of the present disclosure, guide pins **36** are formed of a metal or a metal alloy such as copper, aluminum, stainless steel, or the like. Since metal has high machining accuracy, the alignment accuracy using guide pins **36** is high, and guide pins **36** may tightly fit the corresponding pin holes.

[0022] FIG. **1F** illustrates a top view of the grooves **32** in package component **26** in accordance with some embodiments. The lengthwise direction of grooves **32** is parallel to edges of the package component **26**. When viewed from the top of package component **26**, the grooves **32** extend to the right edge of package component **26**, and may extend to the left edge. Alternatively, the left ends of grooves **32** may be spaced apart from the left edge of package component **26**. Guide pins **36** include some portions in grooves **32**, and further comprise some portions extending beyond the right edge of package component **26**. In accordance with some embodiments of the present disclosure, guide pins **36** have tapered right ends such as rounded right end portions, with the parts of guide pins **36** in grooves **32** having a uniform width and a uniform diameter.

[0023] Referring again to FIGS. **1A-1D**, detachable fiber array unit **40** is formed in accordance with some embodiments. Detachable fiber array unit **40** is configured to be inserted into, and pulled out of, ferrule receptacle **28**. Detachable fiber array unit **40** includes ferrule **42**, ferrule housing **44**, optical fibers **46**, spring push **48**, and features **50**. Detachable fiber array unit **40** is configured to be inserted into the cavity **30** of the ferrule receptacle **28**. In accordance with some embodiments of the present disclosure, ferrule housing **44** has the function of holding and securing a part of the ferrule **42** and a part of the optical fibers **46** as an integrated component.

[0024] The ferrule **42** and ferrule housing **44** may be separate components manufactured separately and then assembled together as the integral component. In accordance with alternative embodiments, the ferrule **42** and ferrule housing **44** may be formed as an integrated unit and formed of a same material, for example, in a same molding process. Throughout the description, ferrule **42** and ferrule housing **44** are collectively referred to as ferrule unit **43**, and may not be shown separately in subsequent figures.

[0025] Features **50** are optically coupled to optical fibers **46**. Features **50** are capable of transferring light, and may be lens array or fiber protrusions, which may (or may not) have the ability of

focusing light. Features **50** may have different pitches than optical fibers **46**, and hence are used as a converting unit for converting pitches. Features **50** may have flat or curved end faces, which may or may not have the capability of focusing light. The pitch of features **50** can be the same as or different from the pitch of optical fibers **46**. There may be a plurality of features **50** arranged as an array.

[0026] In accordance with some embodiments, each of the plurality of features **50** may be connected to, and configured to receive optical signals from, a respective one of the plurality of optical fibers **46**. In accordance with alternative embodiments, each of the plurality of features **50** is configured to send optical signals to a respective one of the plurality of optical fibers **46**. In subsequently discussed examples, it is assumed that optical signals are received into package component **26** from optical fibers **46** through features **50**, while the optical signals may also be transmitted in an inversed direction also.

[0027] Detachable fiber array unit **40** may include spring push **48** attached to the ferrule unit **43**. The attachment may be through screwing, soldering, clipping, or the like. Spring push **48** is flexible, and may be used for securing the detachable fiber array unit **40** in cavity **30**. In accordance with some embodiments of the present disclosure, spring push **48** is formed of a metallic material such as copper, stainless steel, or the like. By forming spring push **48** using metal, the machining accuracy is improved, and spring push **48** have high accuracy for aligning the detachable fiber array unit **40** to cavity **30**. As a comparison, if spring push **48** is formed using other materials such as plastics, the machining accuracy will be low, and the alignment accuracy is degraded.

[0028] FIGS. **1A** and **1B** illustrate a top view and a cross-sectional view, respectively, of photonic receiving unit **10** and detachable fiber array unit **40** in accordance with some embodiments. In FIGS. **1A** and **1B** and subsequent figures, X-direction, Y-direction, and Z-direction are marked. The X-direction is the direction in which detachable fiber array unit **40** is inserted into and pulled out of cavity **30**. The Y-direction is the direction perpendicular to the X-direction, and is parallel to the major top surfaces of package components **26** and **28**. The Z-direction is the direction perpendicular to the major top surfaces of package components **26** and **28**. In addition, the subsequently illustrated figures illustrate some features as being semi-transparent in order to show the features inside these features, while these features may actually be formed of opaque materials or transparent materials in reality.

[0029] FIG. **1A** illustrates that guide pins **36** protrude out of the right edge of package component **26**. In addition, package component **26** may include features **50'**, which are parts of an edge coupler (not shown) of package component **26**. The right ends of features **50'** may protrude out of, or may be flush with, the right edge of the main portion of package component **26**. The number and the positions of features **50'** are designed according to the arrangement of features **50** (in detachable fiber array unit **40**) with a one-to-one correspondence.

[0030] FIG. **1A** also illustrates that package component **26** extends from left side into a left part of ferrule receptacle **28**. Guide pins **36** extend into and are exposed to the cavity **30** of the ferrule receptacle **28**. Also, the end portions of features **50'** are exposed to, and may possibly extend into, the cavity **30** of the ferrule receptacle **28**.

[0031] FIG. **1A** also illustrates detachable fiber array unit **40**, which includes ferrule unit **43** and optical fibers **46** attached to the ferrule unit **43**. In ferrule unit **43**, pin holes **36'** are formed. The formation process may include drilling, molding, or the like. The size of pin holes **36'** is designed to tightly fit guide pins **36**, so that when guide pins **36** are inserted into pin holes **36'**, guide pins **36** are not movable in pin holes **36'** in the Y-direction and Z-direction.

[0032] In accordance with some embodiments of the present disclosure, spring push **48** is attached to ferrule unit **43**. Spring push **48** includes body **48A**, and buckles **48B** attached to the opposite sides of body **48A**. The body **48A** may be used to attach to ferrule unit **43**. The buckles **48B** are flexible, and may move slightly toward body **48A** when being squeezed in the Y-direction.

[0033] In accordance with some embodiments of the present disclosure, the buckles **48B** comprise

protruding portions **48BP**. The protruding portions **48BP** have slanted surfaces facing toward ferrule receptacle **28**, which slanted surfaces help to squeeze protruding portions **48BP** when protruding portions **48BP** are forced into cavity **30**. The protruding portions **48BP** further have slanted surface facing away from ferrule receptacle **28**, which slanted surfaces help to squeeze protruding portions **48BP** when protruding portions **48BP** are pulled out of cavity **30**.

[0034] FIG. **1B** illustrates a cross-sectional view, which shows that support element **52** is attached to the package component **20** through adhesive **54**. Support element **52** supports the right portion of package component **26**. Ferrule receptacle **28** may also be attached to the package component **20** through adhesive **54**. As shown in the cross-sectional view, spring push **48** is secured on ferrule housing **43**, for example, through an adhesive, a screw, or the like.

[0035] FIGS. **1C** and **1D** illustrate a top view and a cross-sectional view of photonic receiving unit **10** and detachable fiber array unit **40**, respectively, after detachable fiber array unit **40** has been inserted into ferrule receptacle **28**. The front end of detachable fiber array unit **40** is inserted into cavity **30** (refer to FIG. **1C**). In accordance with some embodiments of the present disclosure, features **50** of the detachable fiber array unit **40** are aligned to the respective features **50'** of the photonic receiving unit **10**, so that features **50** may be optically and signally coupled to the features **50'**.

[0036] In accordance with some embodiments of the present disclosure, guide pins **36** are inserted into pin holes **36'**. Since the guide pins **36** tightly fit pin holes **36'**, the relative position of the detachable fiber array unit **40** and photonic receiving unit **10** are fixed, which leads to the accurate alignment of features **50** to features **50'**.

[0037] Referring to FIG. **1C**, during the insertion process, the buckles **48B** of the spring push **48** are squeezed by the inner sidewalls of ferrule receptacle **28**, and the detachable fiber array unit **40** is forced into cavity **30**. Furthermore, the ferrule receptacle **28** may have recesses **48BP'** (FIG. **1C**) that are recessed from the inner sidewalls of ferrule receptacle **28** into the sidewall portions of the ferrule receptacle **28**. When the detachable fiber array unit **40** is inserted into the desirable position, the protruding portions **48BP** of buckles **48B**, which protruding portions protrude in the Y-direction (FIG. **1A**) will be seated inside the recesses **48BP'** of the ferrule receptacle **28**.

[0038] In accordance with some embodiments of the present disclosure, after the detachable fiber array unit **40** has been inserted into ferrule receptacle **28** and fixed in position, features **50** may be in physical contact with the respective features **50'**. In accordance with alternative embodiments, after the detachable fiber array unit **40** has been inserted into ferrule receptacle **28** and fixed in position, features **50** are physically spaced apart from, closely located to, and are optically coupled to, the respective features **50'**, so that optical signals may be transmitted between features **50** and features **50'**.

[0039] In accordance with some embodiments of the present disclosure, optical signals may be received from optical fibers **46**, transmitted through features **50**, and optically coupled to features **50'** in package component **26**. The optical signals may be processed and possibly converted to electrical signals in the package component **26**, or transmitted optically to package component **24**, and are converted to electrical signals in package component **24**. Signals may also be transmitted in an inversed direction. For example, electrical signals may be converted as optical signals in package component **24** and transmitted to package component **26**. The electrical signals may also be transmitted to package component **26** and converted to optical signals in package component **26**. The optical signals are optically coupled from features **50'** to features **50**, and then transmitted to optical fibers **46**.

[0040] As shown in FIGS. **1C** and **1D**, the alignment of detachable fiber array unit **40** to photonic receiving unit **10** is achieved by two mechanisms. Guide pins **36** and pin holes **36'** are used to limit the relative movement of detachable fiber array unit **40** from ferrule receptacle **28** in all of the X-direction, Y-direction and the Z-direction. Spring push **48** may be used to limit the relative movement of detachable fiber array unit **40** from ferrule receptacle **28** in the Y-direction. The

buckles **48BP**, when being located in the recesses **48BP'** (FIG. 1C) of ferrule receptacle **28**, may also have the function of limiting the relative movement of detachable fiber array unit **40** from ferrule receptacle **28** in the X-direction, Y-direction, and Z-direction.

[0041] As may be realized, detachable fiber array unit **40** may also be pulled out of ferrule receptacle **28**, for example, by squeezing buckles **48B** of the spring push **48**, so that the protruding portions **48BP** may be slightly pushed out of the recesses **48BP'** in which protruding portions **48BP** are seated. In addition, the protruding portions **48BP** have slanted sidewalls facing the X-direction, so that the pulling force also helps to squeeze buckles **48B**, allowing buckles **48B** to be forced out of cavity **30**.

[0042] FIGS. 2A, 2B, 2C, and 2D illustrate the photonic receiving unit **10** and the detachable fiber array unit **40** in accordance with alternative embodiments. Unless specified otherwise, the materials, the structures, and the formation processes of the components in these embodiments are essentially the same as the like components denoted by like reference numerals in the preceding embodiments. The details regarding the materials, the structures, and the formation processes provided in each of the embodiments throughout the description may be applied to any other embodiment whenever applicable.

[0043] FIGS. 2A and 2B illustrate a top view and a cross-sectional view of detachable fiber array unit **40** and photonic receiving unit **10** in accordance with some embodiments of the present disclosure. The views are obtained before the detachable fiber array unit **40** is inserted into the ferrule receptacle **28**. The top view is obtained from a plane cross-section parallel to the plane defined by the X-direction and the Y-direction, and the plane cross-section cuts through cavity **30**.

[0044] In accordance with some embodiments of the present disclosure, guide pins **36** are attached to ferrule receptacle **28**, rather than being fixed in the grooves of the package component **26**. In accordance with some embodiments of the present disclosure, guide pins **36** are formed of a metal or a metal alloy such as copper, aluminum, stainless steel or the like. Since metal has high machining accuracy, the alignment accuracy using metal guide pins **36** is high.

[0045] In accordance with alternative embodiments, guide pins **36** are formed of a same material as the body of ferrule receptacle **28**, so that guide pins **36** and the body of ferrule receptacle **28** may be formed in a same manufacturing process, and the cost of attaching the separately manufactured guide pins **36** and the body of ferrule receptacle **28** is saved. In accordance with these embodiments, guide pins **36** and the body of ferrule receptacle **28** may be formed of a metal such as copper, aluminum, stainless steel, or the like, or another material such as a plastic, a polymer, a resin, an epoxy, a ceramic, or the like.

[0046] Pin holes **36'** are also formed in the detachable fiber array unit **40**, for example, through drilling, molding, or the like. Alternatively, pin holes **36'** are formed as an integrated part of ferrule housing **43** in the molding process for forming ferrule housing **43**.

[0047] In addition, alignment features **62** are formed inside cavity **30** of the ferrule receptacle **28**. In accordance with some embodiments of the present disclosure, the alignment features **62** have tilted surfaces **62SS**, which are not parallel to the X-direction. A tilt angle θ_1 formed between the tilted surface **62SS** and the X-direction may be in the range between (and greater than) 0 degrees and about 10 degrees. In accordance with some embodiments, alignment features **62** are formed of a material different from the material of the body of ferrule receptacle **28**. Alignment features **62** may be pre-formed, and then placed inside cavity **30**. In accordance with alternative embodiments, alignment features **62** are formed of a same material as the body of ferrule receptacle **28**, so that alignment features **62** and the body of ferrule receptacle **28** may be formed in a same manufacturing process, and are formed as an integral component, for example, in the same molding process.

[0048] Referring to FIGS. 2C and 2D, which illustrate a top view and a cross-sectional view, respectively, the detachable fiber array unit **40** as shown in FIGS. 2A and 2B is inserted into ferrule receptacle **28**. Features **50** are either in physical contact with features **50'**, or may be spaced apart

but are located close to features **50'**, so that features **50** are optically coupled to features **50'**. The spring push **48** (FIG. 2B) is forced to be flat when the detachable fiber array unit **40** is inserted into ferrule receptacle **28**, and may provide a force (in the Z-direction) for tightening the attachment of detachable fiber array unit **40** to the ferrule receptacle **28**.

[0049] In the insertion process, guide pins **36** are aligned to and inserted into pin holes **36'**, so that features **50** may be aligned to features **50'** accurately. In addition, the alignment features **62** may help the alignment. The tilted surfaces **62SS** allow the entrance of cavity **30** to be slightly larger than the size of the detachable fiber array unit **40**, so that the insertion is easy. With the detachable fiber array unit **40** being pushed into cavity **30**, the tilted surfaces **62SS** guide the detachable fiber array unit **40** to the center (in the top view in FIG. 2C) of cavity **30**, and also guide the pin holes **36'** to be aligned to guide pins **36**.

[0050] FIGS. 3A, 3B, 3C, and 3D illustrate the views in accordance with some embodiments. FIGS. 3A and 3B illustrate a top view and a cross-sectional view of detachable fiber array unit **40** and photonic receiving unit **10** in accordance with some embodiments of the present disclosure. The views are obtained before the detachable fiber array unit **40** is inserted into the ferrule receptacle **28**. The fiber array unit **40** and photonic receiving unit **10** in accordance with these embodiments are essentially the same as in the embodiments in FIGS. 2A and 2B, except that alignment rail **43AR** (in detachable fiber array unit **40**) and the corresponding rail recesses are further formed as alignment features. The rail recesses are not illustrated, and are formed in ferrule receptacle **28**, with the shapes and sizes of the rail recesses being configured to allow the alignment rail **43AR** to tightly fit therein and to slide therein.

[0051] Referring to FIG. 3A, alignment rails **43AR** are formed as the side protruding portions of the ferrule housing **43**, and protrude out of (in the Y-directions) the main body of the ferrule housing **43**. As shown in FIG. 3B, alignment rails **43AR** may have slanted top surfaces **43AR-TS**, which are slightly offset from the X-Y plane, which is defined by the X-direction and the Y-direction. A tilt angle **02** formed between the slant to surface **43AR-TS** and the X-Y plane may be in the range between (and greater than) 0 degrees and about 10 degrees.

[0052] FIG. 3B further illustrates spring push **48** in accordance with alternative embodiments. The spring push **48** is flexible in the Z-direction, and has portions raised higher (with a gap underneath) than the top surface of ferrule housing **43**.

[0053] It is appreciated that although alignment rails **43AR** are parts of the detachable fiber array unit **40**, and rail recesses **43AR'** are in the ferrule receptacle **28** in accordance with some embodiments, the locations may be switched. In accordance with alternative embodiments, the alignment rails **43AR** are parts of the ferrule receptacle **28** and in the cavity **30**, and rail recesses **43AR'** are in the detachable fiber array unit **40**.

[0054] In accordance with some embodiments of the present disclosure, when the detachable fiber array unit **40** is inserted into the ferrule receptacle **28**, alignment rails **43AR** slide in rail recesses **43AR'**. Since the front ends of alignment rails **43AR** are smaller than the back ends, it is easy to insert alignment rails **43AR** into rail recesses **43AR'**. The back ends of alignment rails **43AR** have the same height as the height of the back ends of alignment recesses **43AR'**, so that when the detachable fiber array unit **40** reaches the desirable position, the back ends of alignment rails **43AR** tightly fit the entrance portions of rail recesses **43AR'**.

[0055] FIGS. 3C and 3D illustrate the package component **26** and a part of the ferrule receptacle **28** in accordance with some embodiments. FIG. 3C illustrates the structure before package component **26** is attached to the ferrule receptacle **28**. FIG. 3D illustrates the structure after package component **26** is attached to the ferrule receptacle **28**. It is appreciated that FIGS. 3C and 3D may also represent the perspective views of the embodiments shown in FIGS. 2A, 2B, 2C, and 2D. The rail recesses are not shown. Alignment features **53** are formed on support element **52** for aligning package component **26** to ferrule receptacle **28**.

[0056] It is appreciated that the plurality of alignment features of the present disclosure may be

combined in any combination whenever applicable. For example, FIGS. 4A and 4B illustrate the photonic receiving unit **10** and the detachable fiber array unit **40** in accordance with alternative embodiments. In these embodiments, the guide pins **36** are adopted from the embodiments shown in FIGS. 1A through 1F. Accordingly, the guide pins **36** are placed in the grooves **32** in the package component **26**, and is pressed and secured by top lid **34**. The spring push **48**, on the other hand, is adopted from the embodiments shown in FIGS. 2A through 2D. Accordingly, after the detachable fiber array unit **40** is inserted into the ferrule receptacle **28**, the spring push **48** will be forced to be flat, and will provide a force for fixing the relative position of the detachable fiber array unit **40** in the Z-direction.

[0057] FIGS. 5A and 5B illustrate the photonic receiving unit **10** and the detachable fiber array unit **40** in accordance with alternative embodiments. In these embodiments, the guide pins **36** are adopted from the embodiments shown in FIGS. 2A through 2D (and FIGS. 3A and 3B).

Accordingly, the guide pins **36** are attached to ferrule receptacle **28**. The spring push **48**, on the other hand, is adopted from the embodiments shown in FIGS. 1A through 1F. Accordingly, during the insertion of the detachable fiber array unit **40** into the ferrule receptacle **28**, the spring push **48** is squeezed in the Y-direction, and provides a force for fixing the relative position of the detachable fiber array unit **40** in the Y-direction. The spring push **48**, with protruding portions protruding in the Y-directions, also has the function of holding the detachable fiber array unit **40** in ferrule receptacle **28**.

[0058] FIGS. 6A and 6B illustrate the photonic receiving unit **10** and the detachable fiber array unit **40** in accordance with yet alternative embodiments. In these embodiments, both of the guide pins **36** and the spring push **48** are adopted from the embodiments shown in FIGS. 1A through 1F.

Accordingly, the guide pins **36** are placed in the grooves **32** in the package component **26**, and are pushed and secured by top lid **34**. During the insertion of the detachable fiber array unit **40** into the ferrule receptacle **28**, the spring push **48** is squeezed in the Y-direction, and provides a force for fixing the relative position of the detachable fiber array unit **40** in the Y-direction. The spring push **48**, with protruding portions **48BP** protruding in the Y-directions, also have the function of holding the detachable fiber array unit **40** in ferrule receptacle **28**.

[0059] The embodiment as shown in FIGS. 6A and 6B also include the alignment features **62** as shown in FIGS. 2A through 2D (and FIGS. 3A and 3B), which are used for guiding the insertion of the detachable fiber array unit **40** into the ferrule receptacle **28**.

[0060] The embodiments of the present disclosure have some advantageous features. By forming alignment features such as guide pins, spring pushes, alignment features with tilted surfaces, alignment rails, etc., the alignment of the detachable fiber array units to ferrule receptacles is improved. Furthermore, metal may be used for forming alignment features. Due to the high machining accuracy of metal over other materials, the accuracy of the alignment is improved.

[0061] In accordance with some embodiments of the present disclosure, a structure comprises a photonic receiving unit comprising a ferrule receptacle; a die comprising a first waveguide, wherein a part of the first waveguide is in a cavity of the ferrule receptacle; and a guide pin at least partially in the cavity of the ferrule receptacle; and a detachable fiber array unit comprising a ferrule housing; an optical fiber with a part in the ferrule housing; and a second waveguide optically coupled to the optical fiber, wherein an end of the second waveguide is on a side of the ferrule housing; and a pin hole in the ferrule housing, wherein the detachable fiber array unit is configured to be capable of being inserted into the cavity of the ferrule receptacle, and wherein when the detachable fiber array unit is inserted in the cavity, the guide pin comprises a portion in the pin hole, and the second waveguide is optically coupled to the first waveguide.

[0062] In an embodiment, the guide pin comprises a first portion in a groove of the die, and a second portion protruding beyond an edge of the die, with the second portion being in the cavity. In an embodiment, the photonic receiving unit further comprises an additional guide pin, and the detachable fiber array unit further comprises an additional pin hole, and wherein when the

detachable fiber array unit is inserted in the cavity, the additional guide pin comprises a portion in the additional pin hole. In an embodiment, the guide pin is attached to the ferrule receptacle to form an integral component.

[0063] In an embodiment, the detachable fiber array unit further comprises a spring push. In an embodiment, the spring push comprises a metal. In an embodiment, the ferrule receptacle further comprises two alignment features in the cavity, wherein the two alignment features comprise tilted surfaces facing the cavity and opposite to each other, and the detachable fiber array unit is configured to be limited by the tilted surfaces when inserted into the ferrule receptacle. In an embodiment, the guide pin comprises a metal.

[0064] In an embodiment, the ferrule receptacle further comprises a rail recess, and the ferrule housing further comprises a rail on an edge of the ferrule housing, wherein the rail is configured to slide in the rail recess when the detachable fiber array unit is inserted into the ferrule receptacle. In an embodiment, the photonic receiving unit comprises a photonic die and an electronic die, with the die being one of the photonic die and the electronic die. In an embodiment, the structure further comprises a package substrate, wherein the die is over and electrically coupling to the package substrate.

[0065] In accordance with some embodiments of the present disclosure, a structure comprises a first device die comprising a substrate; a guide pin, wherein a portion of the guide pin comprises a part in a groove of the substrate; a ferrule receptacle, wherein the first device die and the guide pin comprise portions in a cavity of the ferrule receptacle; and a detachable fiber array unit comprising a ferrule housing comprising a pin hole, wherein the detachable fiber array unit is configured to be inserted into the ferrule receptacle, and the guide pin is configured to be inserted into the pin hole. In an embodiment, the guide pin comprises a metal, and the substrate comprises a semiconductor substrate.

[0066] In an embodiment, the structure further comprises a second device die overlapped by a first part of the first device die and signally coupled to the first device die; and a package component underlying and electrically coupling to the second device die. In an embodiment, the structure further comprises a support element between the package component and a second part of the first device die. In an embodiment, the structure further comprises an adhesive in the groove and adhering the guide pin to the first device die.

[0067] In accordance with some embodiments of the present disclosure, a structure comprises a ferrule receptacle; a device die extending into a cavity of the ferrule receptacle from a first edge of the ferrule receptacle; a guide pin at least partially in the cavity; and a detachable fiber array unit extending into the cavity from a second edge of the ferrule receptacle, wherein the first edge and the second edge are opposite edges of the ferrule receptacle, and wherein the guide pin comprises a portion in a pin hole of the detachable fiber array unit. In an embodiment, the guide pin is a discrete guide pin that is physically detachable from the ferrule receptacle. In an embodiment, the guide pin is an integral part of the ferrule receptacle. In an embodiment, the detachable fiber array unit further comprises a spring push comprising a part in the cavity.

[0068] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

- 1.** A structure comprising: a photonic receiving unit comprising: a ferrule receptacle; a die comprising a first waveguide, wherein a part of the first waveguide is in a cavity of the ferrule receptacle; and a guide pin at least partially in the cavity of the ferrule receptacle; and a detachable fiber array unit comprising: a ferrule housing; an optical fiber with a part in the ferrule housing; and a second waveguide optically coupled to the optical fiber, wherein an end of the second waveguide is on a side of the ferrule housing; and a pin hole in the ferrule housing, wherein the detachable fiber array unit is configured to be capable of being inserted into the cavity of the ferrule receptacle, and wherein when the detachable fiber array unit is inserted in the cavity, the guide pin comprises a portion in the pin hole, and the second waveguide is optically coupled to the first waveguide.
- 2.** The structure of claim 1, wherein the guide pin comprises a first portion in a groove of the die, and a second portion protruding beyond an edge of the die, with the second portion being in the cavity.
- 3.** The structure of claim 1, wherein the photonic receiving unit further comprises an additional guide pin, and the detachable fiber array unit further comprises an additional pin hole, and wherein when the detachable fiber array unit is inserted in the cavity, the additional guide pin comprises a portion in the additional pin hole.
- 4.** The structure of claim 1, wherein the guide pin is attached to the ferrule receptacle to form an integral component.
- 5.** The structure of claim 1, wherein the detachable fiber array unit further comprises a spring push.
- 6.** The structure of claim 5, wherein the spring push comprises a metal.
- 7.** The structure of claim 1, wherein the ferrule receptacle further comprises two alignment features in the cavity, wherein the two alignment features comprise tilted surfaces facing the cavity and opposite to each other, and the detachable fiber array unit is configured to be limited by the tilted surfaces when inserted into the ferrule receptacle.
- 8.** The structure of claim 1, wherein the guide pin comprises a metal.
- 9.** The structure of claim 1, wherein the ferrule receptacle further comprises a rail recess, and the ferrule housing further comprises a rail on an edge of the ferrule housing, wherein the rail is configured to slide in the rail recess when the detachable fiber array unit is inserted into the ferrule receptacle.
- 10.** The structure of claim 1, wherein the photonic receiving unit comprises a photonic die and an electronic die, with the die being one of the photonic die and the electronic die.
- 11.** The structure of claim 1 further comprising a package substrate, wherein the die is over and electrically coupling to the package substrate.
- 12.** A structure comprising: a first device die comprising a substrate; a guide pin, wherein a portion of the guide pin comprises a part in a groove of the substrate; a ferrule receptacle, wherein the first device die and the guide pin comprise portions in a cavity of the ferrule receptacle; and a detachable fiber array unit comprising: a ferrule housing comprising a pin hole, wherein the detachable fiber array unit is configured to be inserted into the ferrule receptacle, and the guide pin is configured to be inserted into the pin hole.
- 13.** The structure of claim 12, wherein the guide pin comprises a metal, and the substrate comprises a semiconductor substrate.
- 14.** The structure of claim 12 further comprising: a second device die overlapped by a first part of the first device die and signally coupled to the first device die; and a package component underlying and electrically coupling to the second device die.
- 15.** The structure of claim 14 further comprising: a support element between the package component and a second part of the first device die.
- 16.** The structure of claim 12 further comprising an adhesive in the groove and adhering the guide pin to the first device die.

17. A structure comprising: a ferrule receptacle; a device die extending into a cavity of the ferrule receptacle from a first edge of the ferrule receptacle; a guide pin at least partially in the cavity; and a detachable fiber array unit extending into the cavity from a second edge of the ferrule receptacle, wherein the first edge and the second edge are opposite edges of the ferrule receptacle, and wherein the guide pin comprises a portion in a pin hole of the detachable fiber array unit.

18. The structure of claim 17, wherein the guide pin is a discrete guide pin that is physically detachable from the ferrule receptacle.

19. The structure of claim 17, wherein the guide pin is an integral part of the ferrule receptacle.

20. The structure of claim 17, wherein the detachable fiber array unit further comprises a spring push comprising a part in the cavity.
